

AI Use Declaration Form

Course: Introduction to Artificial Intelligence

Lab Title: Lab 1 Part A — Uninformed Search Algorithms

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GitHub Repository Link: [https://github.com/naaaddaitorto-star/Lab 3 Intro To AI Feedforward Networks](https://github.com/naaaddaitorto-star/Lab_3_Intro_To_AI_Feedforward_Networks)

Date Submitted: _____ 28/07/2026 _____

1. AI Use Summary

Question	Student Response
Did you use any AI tool for this lab?	Yes
Estimated percentage of the work influenced by AI	%20Yes
Did you attach evidence of AI use where applicable?	

2. Details of AI Use

Complete the table below for each AI tool used. Add more rows where necessary.

Name of AI Tool Used	Purpose for Using the Tool	Prompt or Instruction Given to the Tool	Part(s) of the Work Influenced by the Tool	How I Verified, Edited, or Improved the AI Output
Claude	I used the AI to understand key coding terms through an interactive section and for debugging. All this was done with the AI in study mode.	• Please help me debug this code. •	1.3,2.3,2.4,2.5	I corrected the code based on the debug output. I also ude the explanation of deravatives. I also used it to get feedback on student reasoning

3. Attachment of AI Output Evidence

Where required, attach evidence of AI use. Tick all that apply. Add more rows where necessary.

Evidence Type	Attached? (Yes, No, N/A)	File Name
AI-generated draft		
Prompt history		
Screenshot of AI interaction	Yes	
Exported AI conversation		
AI-generated code snippet		
Revised version showing student input		

Bug 1 (the immediate error): `torch.max()` returns two things, not one

```
python
prediction = torch.max(logits.data, 1)
```

`torch.max(tensor, dim)` doesn't return just the predicted class — it returns a pair: `(values, indices)` — the actual max values, *and* the index of where that max occurred (which is what we actually want as our "predicted class"). So `prediction` right now is a tuple-like object, not a plain tensor — and comparing a tuple to a tensor with `==` doesn't do what you expect (hence the confusing "bool object has no attribute `sum`" error).

Fix — unpack it, keeping only the indices:

```
python
_, prediction = torch.max(logits.data, 1)
```



The underscore `_` is a Python convention meaning "I don't need this value, throw it away" —

My write up.

1. The CrossEntropyLoss needs to be in the integer form rather than a one hot vector because of the combinations done internally and it is also less costly as the classes grow.

2. With neural networks caling is needed to help the machine learn better prevent machines from viewing it to be more valuable or greater than other features. The feature scalling helps to compute for gradient and if input is too large the training might become unstable. But for random forest such computatuions are not done so the feature scaling is not directly affected.

3. Shuffle = TRUE is used on training Dataloader to help rearrange the data to be used during each epoch. This ensures that the rows selected for each batch is representative of

the datasets with random mix of classes for the model to generalize during training rather than memorising

uestion 1 refinement: be specific about what's being combined — it's log-softmax + negative log-likelihood, and the integer label is used to directly pick out ("index into") the correct class's predicted probability:

CrossEntropyLoss expects an integer label rather than a one-hot vector because internally it combines log-softmax and negative log-likelihood, using the integer to directly index into the predicted probability of the correct class. This is also less memory-costly than storing a full one-hot vector per row, especially as the number of classes grows.

That's a small addition to your existing answer — otherwise you already had the right idea (internal combination + efficiency as classes grow).

Answers 2 and 3 are good as-is — you've correctly connected feature scaling to gradient computation and contrasted it with random forests, and correctly explained shuffle=True's role in preventing memorization of data order.

4. Student Declaration

I declare that:

Declaration Statement	Response (Yes/No)
The submitted work is my own work.	Yes
Any use of AI tools has been clearly declared in this form.	Yes
The prompts, instructions, and AI-generated outputs have been disclosed where applicable.	Yes
I reviewed, tested, edited, and improved any AI-generated content before submission.	Yes
My AI usage does not exceed 25% of the entire work.	20
I understand that undeclared or excessive AI use may be treated as academic misconduct.	Yes

Student Signature: 

Date: _____